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Roll No. ☐☐☐☐☐☐☐☐☐☐☐☐ **Total No. of Pages : 02**

Total No. of Questions : 18

B.Tech.(Electronics & Electrical) (2012 to 2017)
B.Tech.(Electrical & Electronics) (2013 Onwards)
(Sem.–6)

DIGITAL SIGNAL PROCESSING

Subject Code : BTEEE-601
M.Code : 71130

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

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SECTION-A

Write briefly :

1. Write down the basic elements of digital signal processing system?
2. What do you understand by LTI system?
3. Write Laplace and Z transform of sin(t) and cos(t).
4. Define sampling & Nyquist rate?
5. What is Goertzel algorithm? State its applications.
6. Distinguish between FIR and IIR filter.
7. What is the use of pipelining in DSP processor?
8. What is Gibbs phenomena?
9. What are benefits of representing a digital filter in the block diagram form?
10. State the desirable features of DSP processors.

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SECTION-B

11. Find the z-transform of each of the following sequences :

    a) $x(n) = 2^n u(n) + 3\left(\frac{1}{2}\right)^n u(n).$

    b) $x(n) = \cos(n\omega_o)u(n).$

12. Discuss Linear filtering approach for the computation of DFT.

13. Consider the causal linear shift-invariant filter with system function :

$$H(z) = \frac{1 + 0.875z^{-1}}{(1 + 0.2z^{-1} + 0.9z^{-2})(1 - 0.71^{-1})}$$

    Draw a signal flow graph for this system using

    a) Direct form I

    b) Direct form II

14. Draw the architecture of TMS 320C5X.

15. Describe different types of finite word length effects present in Digital filters and ways to rectify them, with the help of examples.

SECTION-C

16. Find the Kaiser window parameters, β and N, to design a low-pass filter with a cut off frequency $\omega_c = \pi/2$, a stop band ripple $\delta_s = 0.002$, and a transition bandwidth no larger than $0.1\ \pi$.

17. Discuss the importance of ROC in digital signal processing.

18. Write a short notes on :

    a. Bilinear Transformation

    b. Circular convolution

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